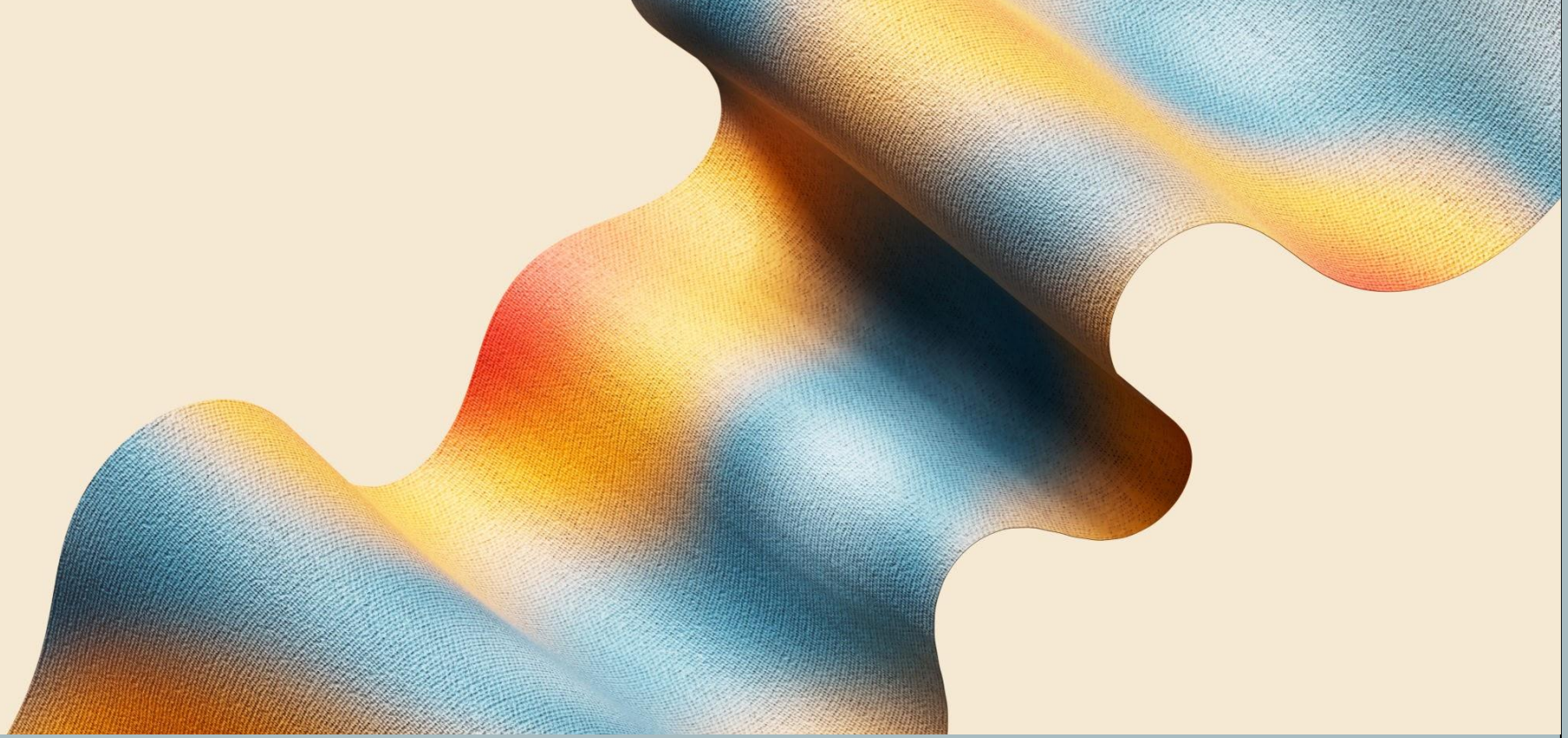


Exploring Latent Color Patterns and Similarity in Bob Ross Paintings

Team members: Kean Zhu, Yihui(Irene) Tang, Yu-Jiun(Janice) Zou, Tzu-Jen(Stephanie) Chen

BA820 · Section B1 · Team 3 · Final Presentation

Unsupervised ML Project

2026.03

Content

01 INTRODUCTION
- Problem, Stakeholders, Why This Matters
- Data & Representation
- Pipeline Overview

02 Method 1 - Clustering

03 Method 2 - Association Rules

04 Final Insights & Limitations

Problem & Stakeholders

Who cares and why This Matters?

Business question

What palette structures appear in Bob Ross paintings?

- one dominant palette style or multiple distinct palette styles?

Stakeholders

- Art learners / instructors: curate learning paths
- Art critics: understand stylistic evolution and recurring themes
- Streaming platforms: tag episodes by palette style for content recommendation

Why this matters

Structured Learning: Enables intentional learning paths from foundational color schemes to more complex compositions

Quantifying Artistic Style: Transforms subjective visual style into measurable structure, enabling systematic comparison

Recommendation: Supports similarity-driven discovery by recommending episodes based on latent palette structure rather than metadata

Data & Representation

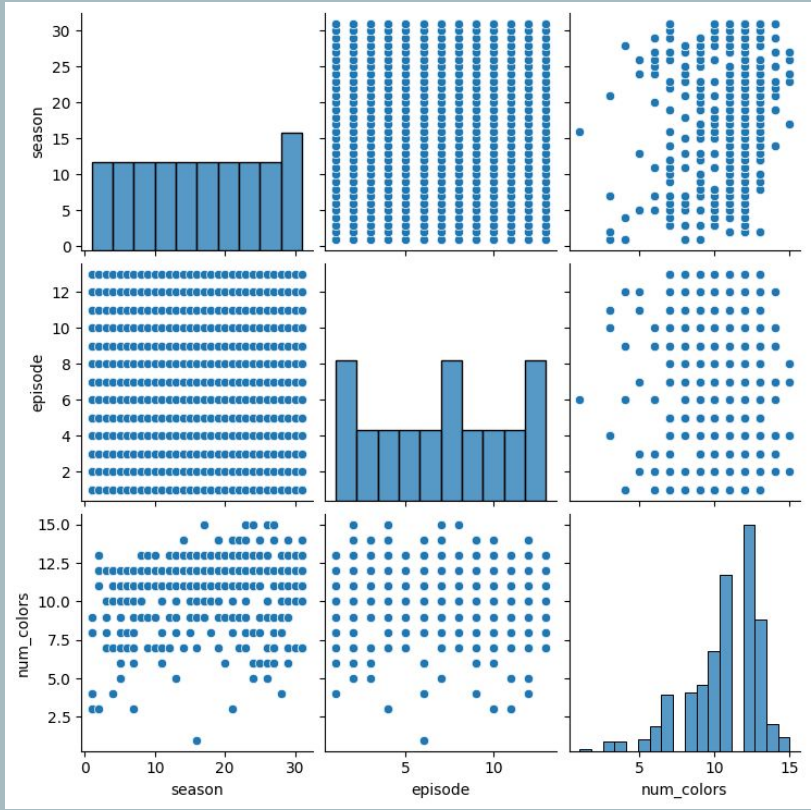
403 paintings (episodes)

- painting_title
- season, episode
- links
- full paints used (binary indicators)

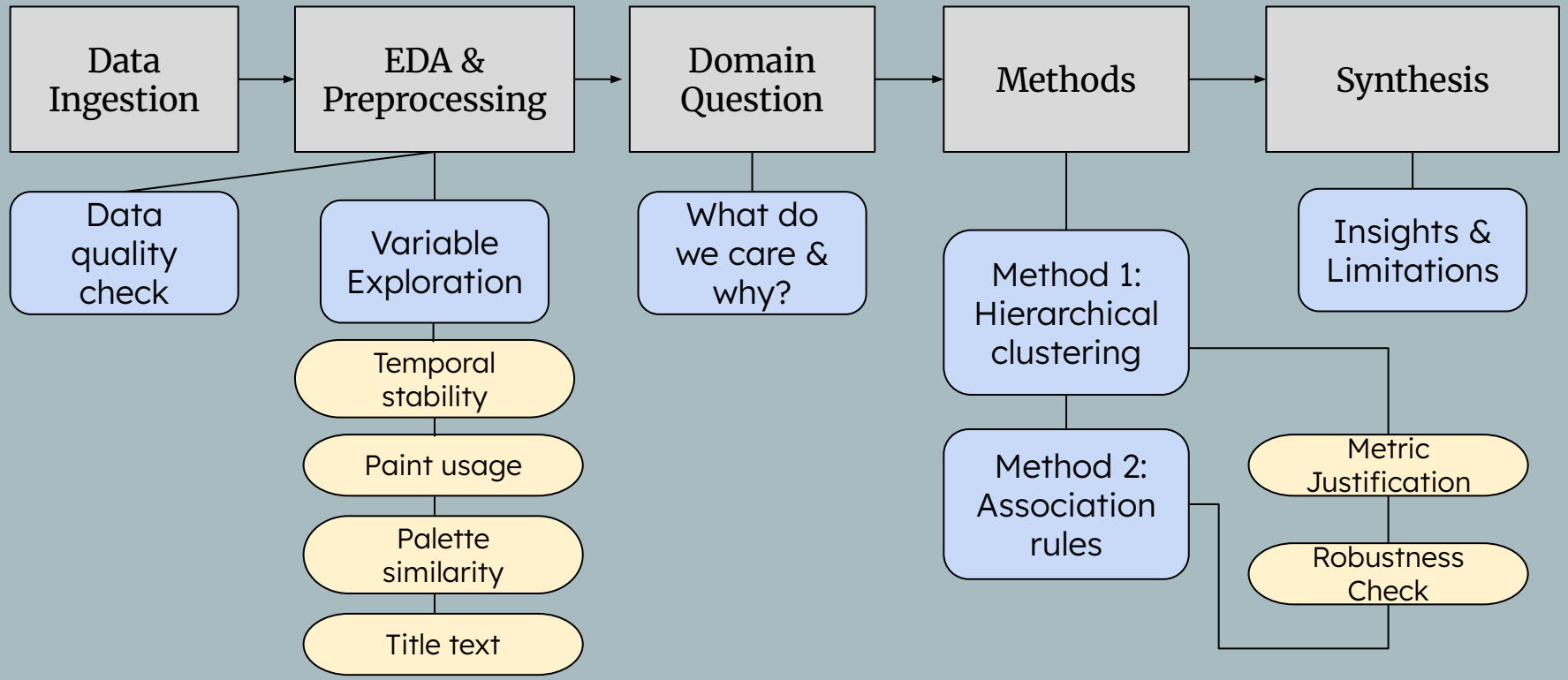
Paint usage is highly imbalanced: a few paints appear in almost every painting.

We represent each painting as a **binary palette vector** (1 = used, 0 = not used), enabling:

- Similarity / nearest-neighbor retrieval
- Grouping and co-usage patterns



Pipeline Overview



Hierarchical Clustering

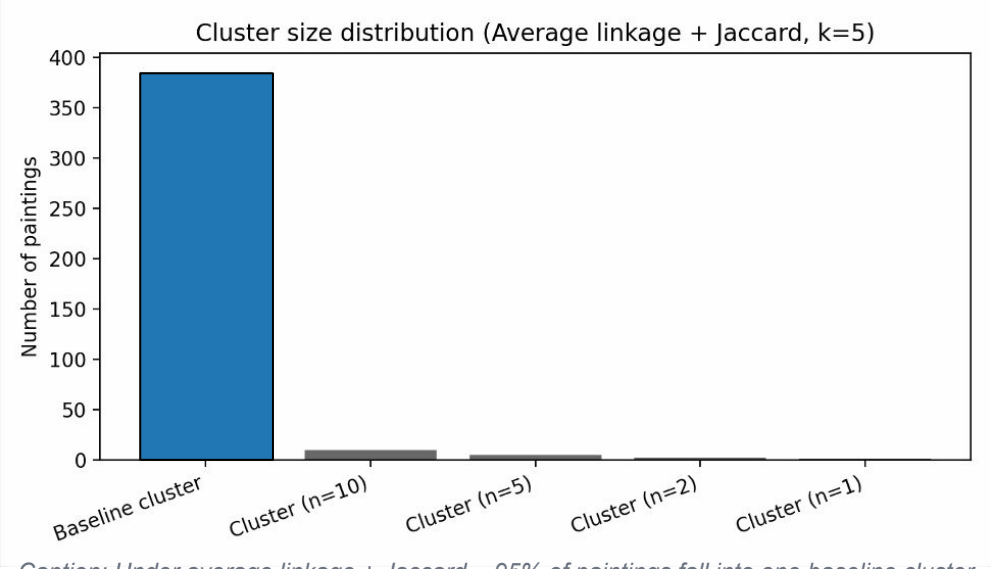
One “standard palette” dominates; small clusters are rare deviations.

- **Goal:** Group paintings by paint sets to find repeatable palette archetypes (baseline vs deviations).
- **Features:** 0/1 paint indicators (each painting = a “basket” of paints).
- **Similarity:** Jaccard distance focuses on shared paints (1s) rather than shared absences (0s).

Example (binary paint usage)

Painting A: 1 0 1 0 0 1 …
Painting B: 1 0 0 0 1 1 …
Painting C: 1 1 1 0 0 0 …

1 = used, 0 = not used



Caption: Under average linkage + Jaccard, ~95% of paintings fall into one baseline cluster.

Hierarchical Clustering

The dominant baseline cluster is stable across robustness checks.

Robustness Check

- Baseline dominance is **stable**
- Fine detail shifts: rare clusters are **sensitive**

Methods Tested for Robustness Check	
Linkage Comparison	Average vs Complete
K sensitivity	k=3/4/5/6
Cluster Interpretation	Baseline vs Deviation

Example: Linkage comparison

Average (Jaccard)	385 / 10 / 5 / 2 / 1
Complete (Jaccard)	331 / 44 / 15 / 13 (4)

- Both linkage methods produce **one dominant cluster** containing the majority of paintings.
  - Differences are concentrated in the small clusters
  - The dominant baseline cluster persists

Association Rules

Examine co-occurrence structures and understand how colors work together within each cluster and what drives cluster variation.

Global vs Cluster Comparison

- Compare full dataset rules with cluster rules
- Shared rules reflect common baseline usage
- Cluster only rules indicate structured, specific pattern

Threshold design

Confidence ≥0.60	Ensure relationship is consistent
Lift ≥ 1.30	Keep strong associations
Max Length 3	Keep it simple and interpretable

Cluster 3 | n=10
min_support=0.300, min_conf=0.60, min_lift=1.30 43 rules, cluster-only = 43

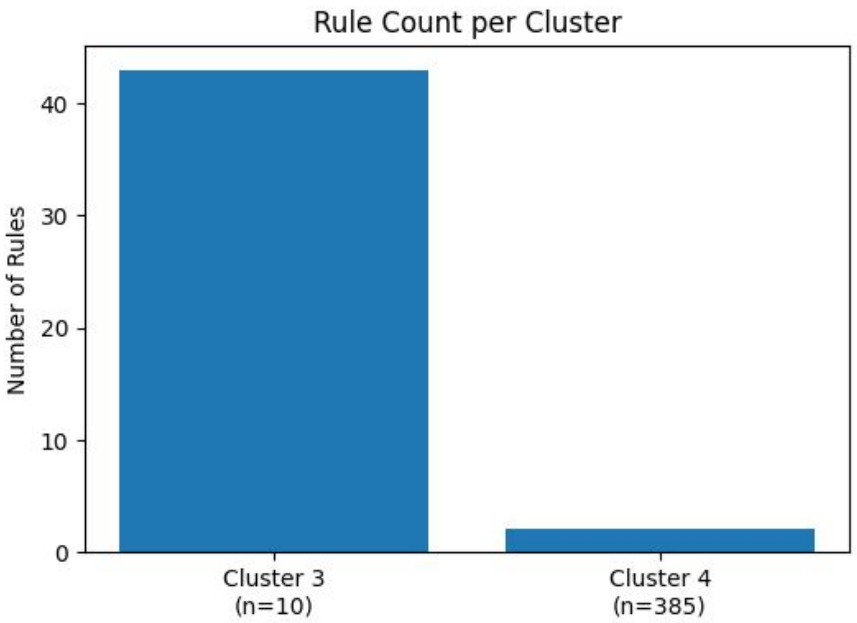
	antecedents	consequents	support	confidence	lift
0	Liquid_Clear	Black_Gesso, Sap_Green	0.3	1.00	2.5
1	Black_Gesso, Sap_Green	Liquid_Clear	0.3	0.75	2.5
2	Liquid_Clear	Black_Gesso	0.3	1.00	2.0
3	Liquid_Clear	Sap_Green	0.3	1.00	2.0
4	Liquid_Clear, Prussian_Blue	Black_Gesso	0.3	1.00	2.0
5	Liquid_Clear	Black_Gesso, Prussian_Blue	0.3	1.00	2.0
6	Liquid_Clear, Sap_Green	Black_Gesso	0.3	1.00	2.0
7	Black_Gesso, Liquid_Clear	Sap_Green	0.3	1.00	2.0
8	Black_Gesso, Midnight_Black	Sap_Green	0.3	1.00	2.0
9	Alizarin_Crimson, Black_Gesso	Sap_Green	0.3	1.00	2.0
10	Liquid_Clear, Prussian_Blue	Sap_Green	0.3	1.00	2.0
11	Liquid_Clear	Prussian_Blue, Sap_Green	0.3	1.00	2.0

Cluster 4 | n=385
min_support=0.100, min_conf=0.60, min_lift=1.30 2 rules, cluster-only = 0

	antecedents	consequents	support	confidence	lift
0	Phthalo_Green	Prussian_Blue, Sap_Green	0.215584	0.715517	1.363733
1	Phthalo_Green	Phthalo_Blue, Prussian_Blue	0.200000	0.663793	1.324147

Association Rules

Differences between clusters come from structured color patterns, not random



Cluster 3 (n=10)

- Generates **43 high-lift rules**
- All rules are cluster-only rules, shows strong structural rigidity

Cluster 4 (baseline cluster, n=385)

- Generates **2 moderate rules**
- Both appear globally, mainly reflects the shared baseline palette

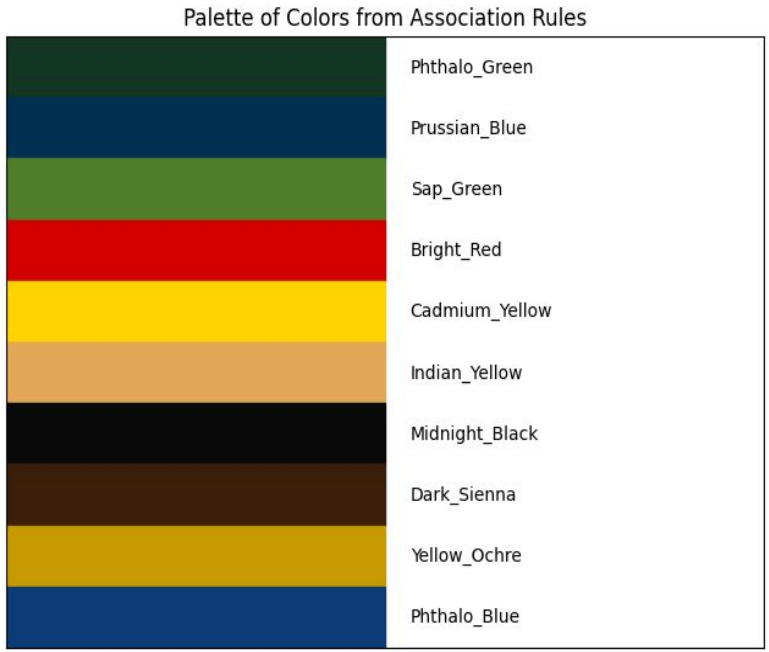
This suggests that differences between clusters come from structured color patterns, not random mixing of paints.

Final Insights

Bob Ross style shows strong underlying structure, not randomly diverse

Key Insights

- Bob Ross relies on one dominant core palette
- Variation comes from small changes in color emphasis
- Similarity between title themes and color grouping patterns is low



Limitations

Theme interpretation may require color proportions, not just binary paint usage

Limitations

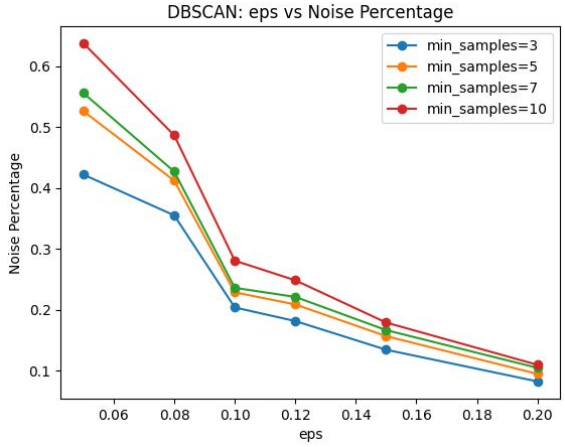
- **We only know whether paints were used, not how they were used**
  Our data only captures structure, but not technique, proportion, blending, or visual intensity
- **Rare clusters are very small**

Cluster	Top title words
Cluster 4 (baseline)	mountain, winter, oval, autumn, lake, cabin
Cluster 3	winter, pot, posies, rays
Cluster 1	winter, grey, sunrise, ocean, fall
Cluster 2	winter, indian, sawscape, girl
Cluster 5	lady, contemplative

Method Tested but Not Selected

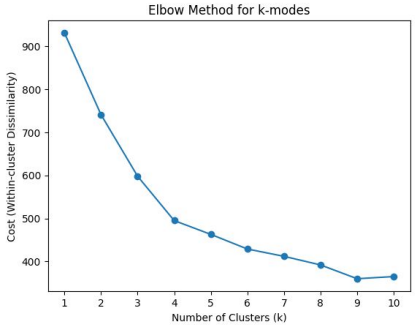
DBSCAN

Across parameter settings, DBSCAN produced one dominant cluster and categorize a lot of paintings to noise points



K-Modes

- Elbow method suggests K=4 as a good cluster structure
- Association rule mining reveals only one cluster produced strong rules under the thresholds



kmodes_cluster	count
0	62
1	165
2	73
3	103

```
Cluster 1 | n=165
min_support=0.100, min_conf=0.60, min_lift=1.30
No rules found under these thresholds.

Cluster 2 | n=73
min_support=0.100, min_conf=0.60, min_lift=1.30
No rules found under these thresholds.

Cluster 3 | n=103
min_support=0.100, min_conf=0.60, min_lift=1.30
No rules found under these thresholds.
```

Thank You!

Q & A

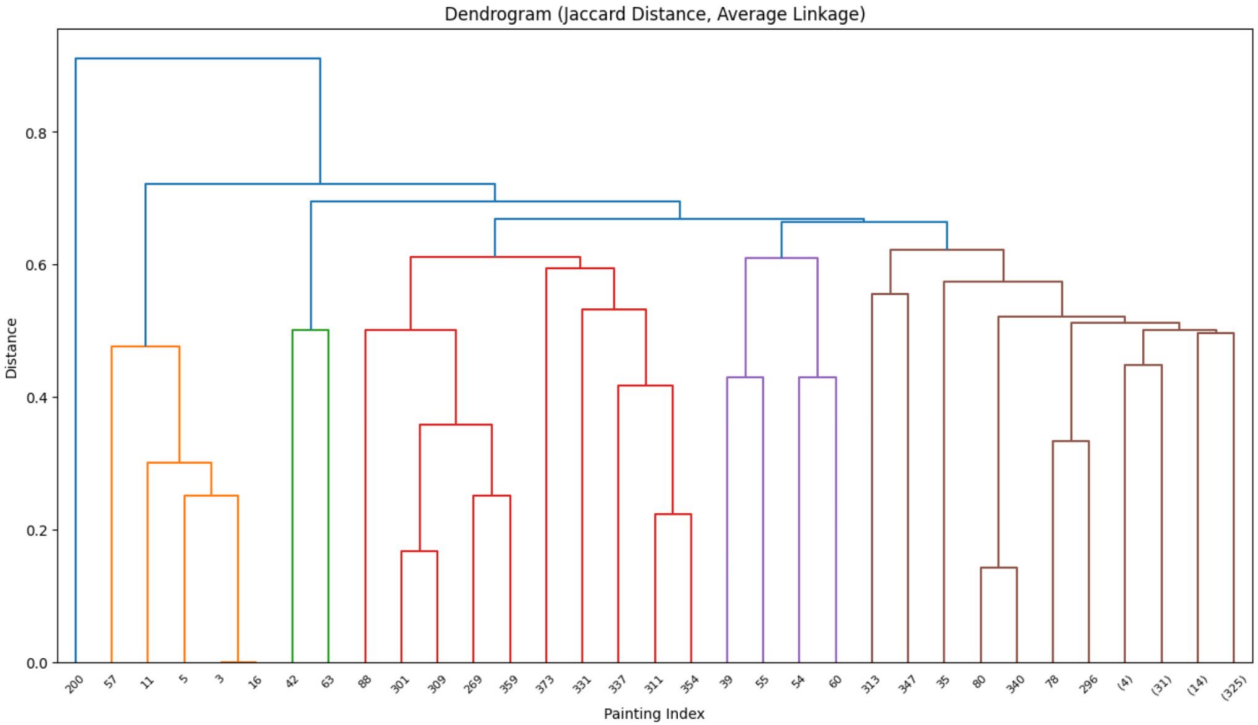
Team 03 members:

Kean Zhu, Yihui(Irene) Tang,

Yu-Jiun(Janice) Zou,

Tzu-Jen(Stephanie) Chen

Appendix



	k	Silhouette Score
0	2	0.597944
1	3	0.498231
2	4	0.470907
3	5	0.466916
4	6	0.455069
5	7	0.431152
6	8	0.424071